



Characterization of Root Architectures and Soil Macropore Networks Under Different Ecosystems Using X-ray CT Scanning in the Qinghai Lake Watershed, NE Qinghai–Tibet Plateau

Zong-chao Li^{1,2} · Xia Hu^{1,3} · Xiao-yan Li^{1,3}

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Abstract

The objectives of this study were to visualise and quantify the soil macropore networks and root architectures within intact soil columns under different ecosystems and to investigate the influences of roots on soil macropore network characteristics. Intact soil columns with diameters of 110 mm and lengths of 400 mm were taken from alpine *Kobresia* meadow, interspace meadow patches of *Potentilla fruticosa* shrubs, *Potentilla fruticosa* shrubs and *Artemisia sphaerocephala* shrubs in the Qinghai Lake watershed of the NE Qinghai–Tibet Plateau. A Philips medical X-ray CT scanner was used to simultaneously visualise and quantify the soil macropore networks and root architectures. The results showed that macropores and taproots were concentrated in the top 0–150 mm. The root volume density of the *P. fruticosa* shrub was greater than that of the interspace meadow patches of *P. fruticosa* shrub and that of the alpine *Kobresia* meadow. The macroporosity and the root volume density of the soils under the *P. fruticosa* shrub patches were significantly greater than those in the soils under the other sites. The root volume density was significantly correlated with the macroporosity for the *A. sphaerocephala* shrubs ($R^2 = 0.88$) and the *P. fruticosa* shrubs ($R^2 = 0.73$). There were significant positive correlations between the volume density, branch density and mean angle of the roots and those of the macropores. Shrub roots invasion played an important role in the formation of macropores. The root volume density was the most important factor for the quantified characteristics and distributions of the macropores.

Keywords X-ray CT · Soil macropore · Root · Quantification · Correlation

1 Introduction

Soil macropores play an important role in soil functioning (Brewer 1965; Perret et al. 1999; Fox et al. 2004; Luo et al. 2008; Dorner et al. 2010) and have a direct effect on water infiltration, subsurface storm flow and solute transport (Keith

and Peter 1982; Sidle et al. 2001; Luo et al. 2010b). Macropores are comprised of large soil voids and pathways, including root channels, cracks, fissures (Keith and Peter 1982; Haling et al. 2013), and earthworm burrows (Capowiez et al. 2016), which can permit the preferential flow of water and nutrients through the soil profile. Roots, water content and soil compaction are important factors of macropore formation (Gebhardt et al. 2009; Hosseini et al. 2018; Hu et al. 2018). Furthermore, soil pore size distribution is an important soil physical property in relation to root development through their effect on mechanical resistance (Hosseini et al. 2016). Some studies have shown that the growth of roots modifies the physical environment by moving soil particles, deforming aggregates and decreasing the number of interaggregate pores (Aravena et al. 2013; Pagenkemper et al. 2013). The presence of soil macropores significantly benefits root growth (Bennie 1991) because continuous macropores can positively affect the flow of water and nutrients to roots (Hatano et al. 1988; Chapman and Miller 2011). In turn, plant roots effectively form biological macropores that can be used as preferential flow pathways,

✉ Xia Hu
huxia@bnu.edu.cn

Zong-chao Li
zongchaoli@mail.bnu.edu.cn

¹ Key Laboratory of Earth Surface Processes and Resource Ecology, Beijing Normal University, Beijing 100875, China

² China National Environmental Monitoring Centre, Beijing 100012, China

³ School of Natural Resources, Faculty of Geographical Science, Beijing Normal University, Beijing, No. 19, Xijiekouwai Street, Haidian District, Beijing 100875, China

and thus, they are a major contributor to preferential infiltration mechanics (Gish et al. 1998; Leonard et al. 2004; Pierret et al. 2007). The rhizosphere and root channels both constitute root-induced macropores that can modify the architectures of soil macropores (Aravena et al. 2011; Devitt and Smith 2002). Furthermore, root-induced macropores are generated mainly within the compact soil layer (Lesturgez et al. 2004; Pfeifer et al. 2015). In addition, growing roots modify the physical properties of the rhizosphere (Aravena et al. 2013), resulting in significant improvements of the macroporosity and intra-aggregate porosity of this compact layer (Lesturgez et al. 2004). However, the effects of root growth on the soil structure, especially on the three-dimensional (3-D) macropore characteristics, remain poorly understood in alpine environments.

It is essential to quantify the physical, chemical and biological functions of the macropore characteristics and to predict their dynamics (Allaire et al. 2002; Capowiez et al. 2016; Luo et al. 2008; Peth et al. 2008). Over the last decade, the X-ray computed tomography (CT) scanning method has been widely employed to research soil structures and particularly to visualise macropore networks (Bottinelli et al. 2016; Iassonov et al. 2009; Perret et al. 1999; Udawatta and Anderson 2008; Zhang et al. 2017). Quantifying the 3-D geometry and topology of a macropore network includes the macroporosity, macropore size distribution, surface area, branch length, continuity, tortuosity and connectivity (Luo et al. 2010a; Perret et al. 1999; Peth et al. 2008; Pierret et al. 2002; Zhang et al. 2017). Furthermore, the non-destructive X-ray CT scanning technique has been applied to analyse the distribution and architectures of root networks (Haling et al. 2013; Mooney et al. 2012). In general, homogeneous sandy soils (Flavel et al. 2012; Mooney et al. 2012; Pierret et al. 1999) have constituted the optimum medium for the non-destructive visualisation (Kaestner et al. 2006; Mairhofer et al. 2013) and quantification of root networks using X-ray CT scanning. At a smaller scale, microfocus X-ray CT scanning allows the precise study of the 3-D rhizosphere in non-homogeneous in situ samples (Aravena et al. 2011; Seigniez et al. 2010). Haling et al. (2013) verified that X-ray CT scanning could be used to visualise and analyse the in situ rhizosphere of root decomposition. In addition, the morphological parameters of biopores and the properties of network skeletons were quantified using X-ray CT to assess the influences of root-induced biopores on pore space architectures (Pagenkemper et al. 2013). Furthermore, Kuka et al. (2013) investigated the soil–root relationship in grassland soils using X-ray CT and indicated that the relationship between the soil structure and the root pattern is related to the intensity of land management (Kuka et al. 2013).

Kobresia meadows and *Potentilla fruticosa* shrubs with mosaic distributions in those meadows are widely distributed throughout the river valleys of the Qinghai Lake watershed of the NE Qinghai–Tibet Plateau. Previous studies have illustrated the ecological function of the alpine shrub ecosystem (He

and Li 2016; Hu et al. 2016; Zhang et al. 2015), and some studies have noted the impacts of shrub roots on the soil structure (Hu et al. 2015; Li et al. 2013). Hu et al. (2016) inferred that the large number of macropores found in the soils under *P. fruticosa* shrubs could be attributed to more substantial root development. However, few studies have attempted to quantify the interactions between soil macropore networks and shrub root architectures, particularly within the Qinghai–Tibet Plateau. Therefore, the objectives of this study were to (i) visualise and quantify the spatial characteristics of the soil macropores and root architecture of a intact soil column and (ii) quantify the correlation between the soil macropores and plant roots in the soils of *P. fruticosa* shrubs and *Artemisia sphaerocephala* shrubs in the Qinghai Lake watershed of the NE Qinghai–Tibet Plateau.

2 Materials and Methods

2.1 Study Sites and Core Samples

The Qinghai Lake watershed is located in the NE Qinghai–Tibet Plateau, which is situated in a cold, semiarid, high-altitude climate zone. Alpine meadow constitutes the main ecosystem in the Qinghai Lake watershed, and it accounts for 55.9% of the total land area (Zhang et al. 2015). Subalpine deciduous shrubs, which exhibit a mosaic distribution throughout the alpine meadow ecosystem, account for 5.35% of the total area. Furthermore, desert accounts for 4.03% of the total land area in the Qinghai Lake watershed. The selected sites consisting of alpine *Kobresia* meadow, *P. fruticosa* shrubs and interspace meadow patches of *P. fruticosa* shrubs (36° 47' 30" N, 100° 47' 6" E, pentagon in Fig. 1) were located along the Shaliu river, which is one of the largest rivers in the Qinghai Lake watershed. At the site of *P. fruticosa* shrubs, the main soil type was peat soil and the main species were *P. fruticosa*, *Carex moorcroftii*, and *P. saundersiana* shrub. The soil type of *A. sphaerocephala* shrubs site (37° 35' 40" N, 100° 0' 26" E, triangle in Fig. 1), which is located in east desert of Qinghai Lake, was homogeneous sandy soil and the *A. sphaerocephala* shrub site was chosen to verify the validity of the X-ray CT scanning method for plant roots.

Twelve undisturbed soil cores (with 3 replicates) were sampled using polyvinyl chloride (PVC) pipes with a diameter of 110 mm and a length of approximately 500 mm. The above-ground parts of the plants were carefully clipped with minimal disturbance prior to the soil sampling. Cylindrical soil cores (with a diameter of 110 mm) were made by cutting into the soil around each sample. Then, we manually pushed the PVC pipes vertically and gradually into the soil. After sampling, both ends of each PVC pipe were packed and sealed. All of the soil cores were scanned using a medical X-ray CT device as quickly as possible after collection.

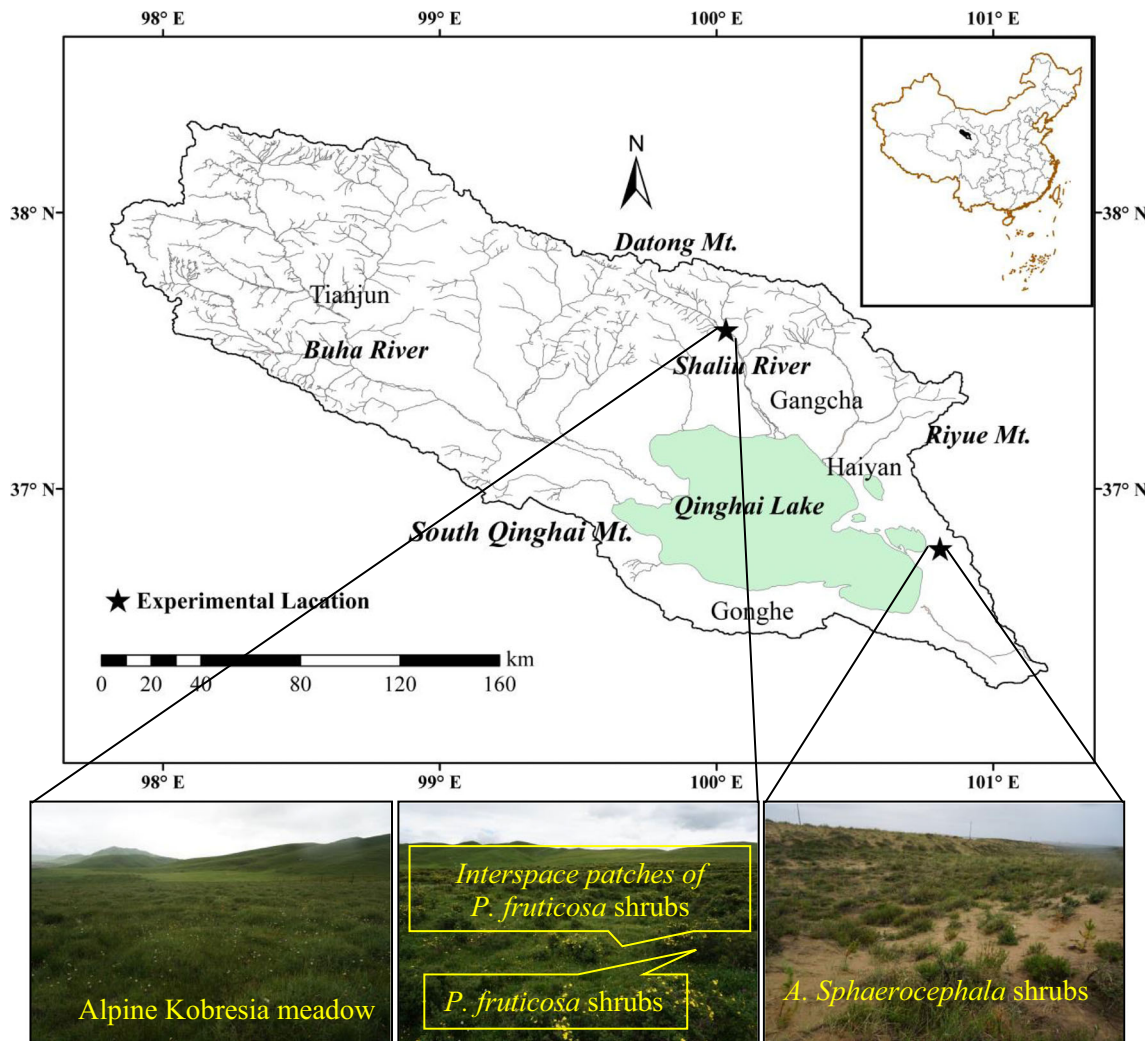


Fig. 1 Study area and sampling sites. The star in Shaliu River represents the sampling area which contains 3 sampling sites (Alpine *Kobresia* meadow, *P. fruticosa* shrubs and interspace meadow patches of *P. fruticosa* shrubs)

After scanning, each soil column was cut into three sections (0–100 mm, 100–200 mm and 200–400 mm), and a soil sample was collected from each section for soil property analysis. The particle size composition was measured using a laser particle analyser (Malvern Mastersizer 2000) equipped with a HydroMu dispersing unit; the analytical range is 0.02–2000 μm . The particle size composition was classified into clay (< 2 μm), silt (2–20 μm) and sand (20–2000 μm) particles according to the international Polish Soil Classification (PSC) system. The organic matter content was measured using the potassium bichromate titrimetric method of Walkley and Black (1934). The total phosphorus and total nitrogen concentrations were measured using the semi-micro Kjeldahl method.

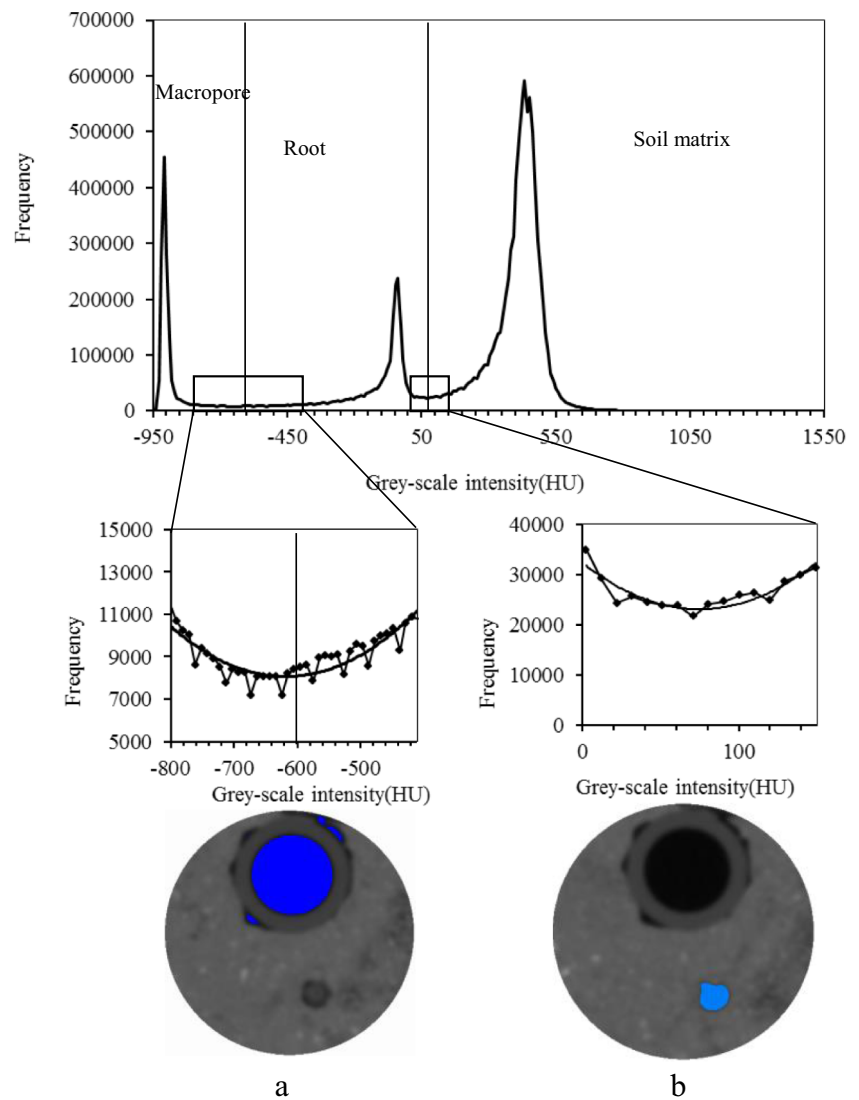
2.2 CT Scanning and Image Analysis

A Philips Brilliance ICT 256 medical scanner (i.e. a 256-slice spiral CT scanner) was used to scan the soil cores at an energy level of 140 kV and 300 mA with a 0.312-mm scanning

interval to provide detailed, low-noise projections. The CT scanner produced 16-bit images with a slice thickness of 0.625 mm and 1024×1024 pixels per slice, thereby producing voxels with a resolution of $0.146 \text{ mm} \times 0.146 \text{ mm} \times 0.312 \text{ mm}$ in the reconstructed image. After scanning and reorientation, approximately 1400 slices were produced for every soil core with a signal-to-noise ratio (SNR) of 2.76.

The slice images were reconstructed and analysed using AVIZO Fire 9.0 (FEI, America). A cylindrical cropping tool was applied to obtain a region of interest (ROI) to remove obvious disturbances caused by the sampling. Three peaks were distinguishable in the intensity histogram of the raw data (Fig. 2). Intensities of about – 600 HU (Fig. 2) were obtained from the intensity histogram by fitting a parabola similar to the method of Katuwal et al. (2015). The calculated porosity using the threshold value – 600 HU and the measured porosity was less than 1% (Li et al. 2013), which verified the method of CT scanning. The roots were segmented from the reconstructed CT data by using the region growing method (Gregory et al. 2003) in AVIZO Fire

Fig. 2 Grey scale intensity (HU) histograms generated using the thresholding method. Grey scale image with the segmented macropores shown in blue (**a**); grey scale image with the segmented roots shown in blue (**b**)



9.0. In brief, an interactive module allowed a human operator to track each root through a stack of CT slices by marking the centre of the root region. We marked only the regions with intensities between -500 HU and 70 HU as the centre of a root.

After segmentation and binarisation, the soil macropore and root networks were visualised. Meanwhile, the macroporosity (volume of all macropores/volume of the ROI), macroporosity of a slice (volume of all macropores on the slice/volume of the slice), root volume density (volume of all roots on the slice/volume of the ROI), root volume density of a slice (volume of all roots on the slice/volume of the slice), volume and surface area of the network are calculated by a module (volume fraction) in Avizo 9.0. A node was defined as an intersection where two branches (the segment in the skeletonisation) were connected. The coordination number is the number of branches that connect to a node. Vertical length was the vertical distance from starting points to ending points of a network. The inclination (angle) was

characterised by an angle between a straight line (from starting points to ending points) and vertical direction. Thus, it was necessary to skeletonise the roots and macropores to be a centreline network composed of dots and lines. The total length, vertical length, branch length, mean tortuosity and angle of each network were quantified by skeletonisation. The overall network number, node number, branch number, mean volume size and coordination number were counted by the statistics module of Avizo 9.0. The surface area density, length density, network number density, node density and branch density were the surface area, length, number of networks, node and branches in a unit volume, respectively. Assuming that all macropores are cylindrical, its mean equivalent radius was calculated by the total volume and total length of the macropore. Meanwhile, the mean tortuosity was calculated as ratio of the total actual macropore length to total straight line distance of all the macropores in a certain volume (Luo et al. 2010a).

2.3 Destructive Quantification of Root Parameters

The homogeneous sandy soil of the *A. sphaerocephala* shrubs, which exhibited a high sand content (97%) and low organic matter content (2.09 g kg^{-1}), was used to validate the X-ray CT scanning method. After scanning, the 3 cores of the *A. sphaerocephala* shrubs were cut into three sections (0–100 mm, 100–200 mm and 200–400 mm) and the roots were collected. The roots of 9 sections were subjected to a manual washing procedure with a 0.25-mm sieve to separate the individual roots from the soil (Kuka et al. 2013). Root volume and length were quantified by the method of a standard root washing with the software WinRhizo 2009 (regent instruments) (Arsenault et al. 1995).

2.4 Statistical Analysis

A comparative data analysis was only applied to the root attributes acquired from the X-ray CT scanning technique and a destructive technique when comparing the total volume and length of the root system due to the lack of available data on very fine roots ($<0.5 \text{ mm}$) within the 3-D images. One-way analysis of variance (ANOVA) was used to detect variations in the differences between the measured root and macropore properties (e.g. the volume density, length density, surface area density, mean size, node density and diameter). Pearson's correlation coefficient (R) was used to test the correlations between the different macropore characteristics and between the root characteristics and macropore characteristics. The statistical analyses were performed using version 13.0 of the SPSS software package.

3 Results

3.1 Validation of the X-ray CT Scanning Technique for Roots in Soil Cores

The homogeneous sandy soil of the *A. sphaerocephala* shrubs was chosen to verify the suitability of the medical X-ray CT

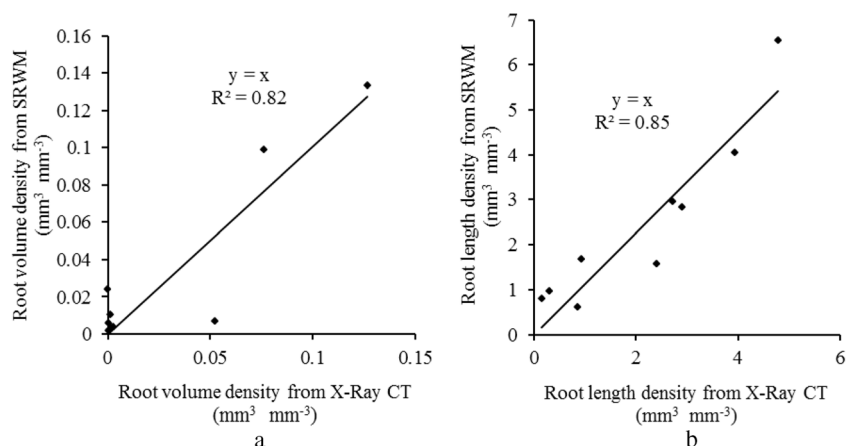
methodology for root evaluation. The root volume density and length density were the ratio of total volume and length of all roots in each section to the volume of the section, respectively. Figure 3 shows the relationship between the root volume density and length density acquired using X-ray CT scanning and those acquired using a standard root washing method (SRWM) with subsequent scanning. The relationship between the two methods with respect to the root volume was evidently strong with a calculated coefficient of determination (R^2) value for $y = x$ of 0.82. There was also a strong relationship between the two methods with respect to the root length with a calculated R^2 value of 0.85 (Fig. 3).

3.2 3-D Visualisation and Quantification of Root Architectures

The 3-D visualisations of the root architectures in the 12 soil columns are shown in Fig. 4, and the variations in the root volume fraction (i.e. the root volume/slice volume) with depth are shown in Fig. 5a–d. The characteristics of the root architectures were distinctly different among the alpine *Kobresia* meadow, the interspace meadow patches of *P. fruticosa* shrubs, the *P. fruticosa* shrubs and the *A. sphaerocephala* shrubs. Similar to the results of Hudek et al. (2017), two main root system types were observed among all of the soil cores: a dominant pole-type root system characterised by a strong horizontal spreading of the roots and a cylindrical-shaped root system.

For the soils under the alpine *Kobresia* meadow, a large number of fine herb roots were observed throughout the entire column (Fig. 4a). Furthermore, the root volume density decreased slightly with depth between the soil depths of 0 and 300 mm (Fig. 5a). For the soils under the interspace meadow patches of *P. fruticosa* shrubs, the number of roots was obviously less than that beneath the alpine *Kobresia* meadow; moreover, most of the roots were concentrated in the surface layer (0–150 mm) (Fig. 5b). However, the roots of the interspace meadow patches of *P. fruticosa* shrubs were coarser and more horizontal than the roots of the alpine *Kobresia* meadow

Fig. 3 Comparison between the X-ray CT scanning technique and the standard root washing method (SRWM) by using the unary linear regression method. **a** Linear regression between the root density obtained from the X-ray CT scanning technique and from the SRWM. **b** Linear regression between the root length density obtained from the X-ray CT scanning technique and from the SRWM



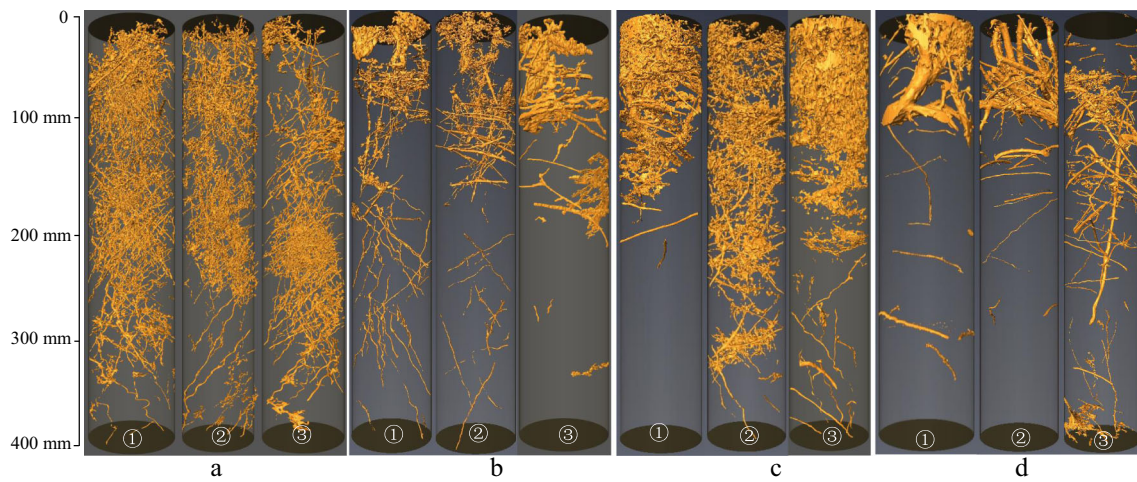


Fig. 4 Three-dimensional visualisation of the root architectures in the soil cores under **a** the alpine *Kobresia* meadow, **b** interspace meadow patches of *P. fruticosa* shrubs, **c** *P. fruticosa* shrubs and **d** *A. sphaerocephala* shrubs

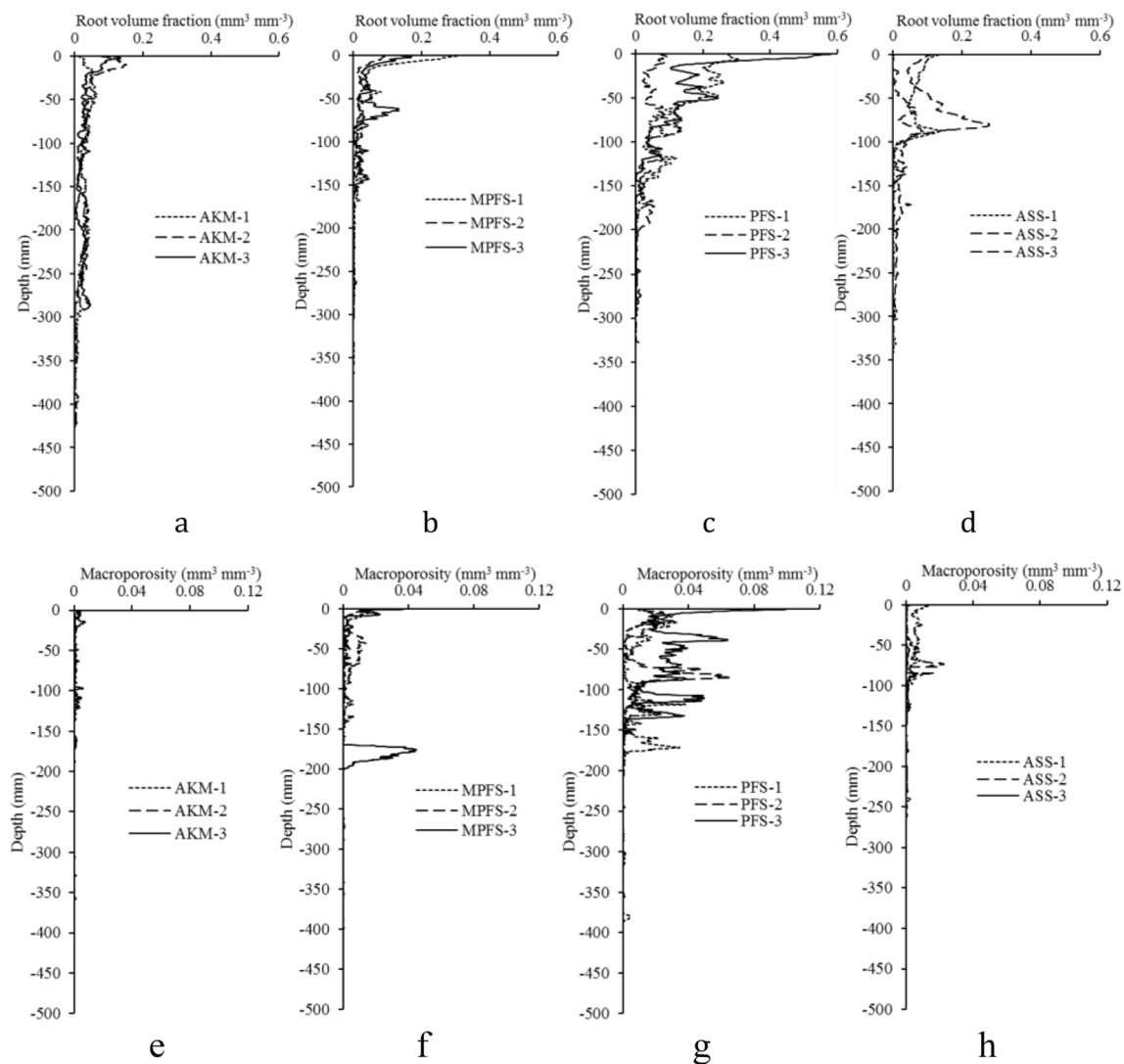


Fig. 5 Distribution of Root volume fraction along the depth of the soil column for the AKM (alpine *Kobresia* meadow) (**a**), MPFS (interspace meadow patches of *P. fruticosa* shrubs) (**b**), PFS (*P. fruticosa* shrubs) (**c**)

and ASS (*A. sphaerocephala* shrubs) (**d**). Distribution of macroporosity along the depth of the soil column for the AKM (**e**), MPFS (**f**), PFS (**g**) and ASS (**h**). Number 1, 2, 3 represent duplicate samples

Table 1 Quantified 3-D characteristics of the root architectures for all of the soil cores as determined from the CT images

	Root volume density ($\text{m}^3 \text{ m}^{-3}$)	Root number density ($\times 10^5$) (no. m^{-3})	Surface area density ($\text{m}^2 \text{ m}^{-3}$)	Length density (km m^{-3})	Node density ($\times 10^5$) (no. m^{-3})	Branch density (no. m^{-3})	Mean root length (mm)	Mean branch length (mm)	Mean root size (mm^3)	Equivalent radius (mm)	Mean coordination number	Tortuosity	Mean angle ($^\circ$)
AKM1	0.0024	0.02	0.0092	24.17	56.56	67.10	7.13	3.60	45.24	0.13	2.37	1.16	51.87
AKM2	0.0021	0.02	0.0800	23.73	58.97	73.96	6.06	3.21	308.34	0.15	2.51	1.17	51.19
AKM3	0.0025	0.02	0.0605	16.90	36.37	45.23	7.51	3.74	393.29	0.17	2.49	1.18	54.07
Mean	0.0023 (± 0.0001) ^a	0.023 (± 0.00)	0.0499 (± 0.0122) ^a	21.60 (± 1.36)	50.63 (± 4.14) ^a	62.10 (± 5.00) ^a	6.9 (± 0.25)	3.52 (± 0.09)	248.96 (± 60.49)	0.15 (± 0.01)	2.46 (± 0.02)	1.17 (± 0.01)	52.38 (± 0.5)
MPFS1	0.0169	0.21	0.0351	9.45	21.17	29.59	19.26	3.19	365.70	0.14	2.80	1.21	57.29
MPFS2	0.0091	0.17	0.0322	11.38	30.84	49.99	24.77	2.28	40.53	0.14	3.24	1.19	53.53
MPFS3	0.0183	0.08	0.0288	6.52	15.74	19.77	11.73	3.30	431.79	0.23	2.69	1.11	69.11
Mean	0.0147 (± 0.0050) ^a	0.15 (± 0.02)	0.032 (± 0.0010) ^a	9.12 (± 0.81)	22.58 (± 2.55) ^a	33.12 (± 5.14) ^a	18.59 (± 2.18)	2.92 (± 0.19)	279.34 (± 69.81)	0.17 (± 0.02)	2.91 (± 0.1)	1.17 (± 0.02)	59.98 (± 2.71)
PFS1	0.0463	0.03	0.0685	26.21	40.44	61.45	30.48	4.26	5585.85	0.26	3.23	2.42	52.16
PFS2	0.0293	0.16	0.0784	25.33	71.76	111.66	22.06	2.27	289.53	0.21	3.11	1.19	54.90
PFS3	0.0593	0.09	0.0885	38.83	64.97	113.22	7.07	3.43	289.09	0.12	3.49	1.55	47.24
Mean	0.045 (± 0.005) ^b	0.09 (± 0.02)	0.0785 (± 0.0033) ^b	30.12 (± 2.52)	59.06 (± 5.49) ^b	95.44 (± 9.82) ^b	19.87 (± 3.95)	3.32 (± 0.33)	2054.82 (± 1019.32)	0.2 (± 0.02)	3.27 (± 0.06)	1.72 (± 0.21)	51.43 (± 1.29)
ASS1	0.0193	0.18	0.0182	4.75	10.15	17.84	6.36	2.66	746.07	0.17	3.52	1.21	44.13
ASS2	0.0269	0.08	0.0253	25.36	4.18	9.74	42.19	26.02	1127.83	0.23	4.66	2.40	52.85
ASS3	0.0136	0.23	0.0265	6.09	12.38	15.57	18.18	3.91	155.90	0.22	2.52	1.18	57.23
Mean	0.0199 (± 0.0022) ^a	0.16 (± 0.03)	0.0233 (± 0.0015) ^a	12.07 (± 3.84)	8.9 (± 1.41) ^a	14.39 (± 1.39) ^a	22.24 (± 6.09)	10.86 (± 4.38)	676.6 (± 163.22)	0.21 (± 0.01)	3.56 (± 0.36)	1.6 (± 0.23)	51.41 (± 2.22)

AKM, MPFS, PFS and ASS represent the samples of alpine *Kobresia* meadow, interspace meadow patches of *P. fruticosa* shrubs, *P. fruticosa* shrubs and *A. spharoccephala* shrubs, respectively. The numbers in parentheses are standard errors of the mean. The letters after the parentheses indicate the significance test of the mean difference among the four groups of soil columns at $p < 0.05$. Mean comparison by using Tukey HSD test

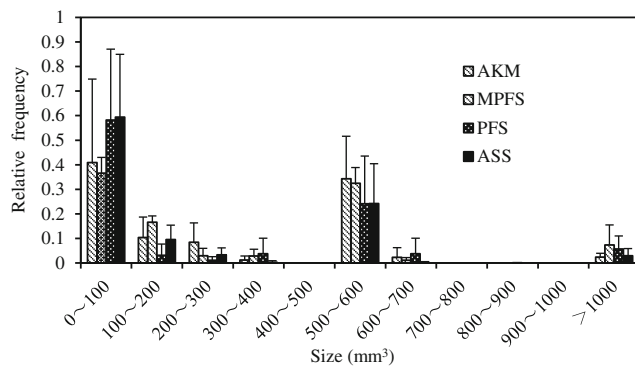


Fig. 6 Cumulative relative frequency distributions of the root size in the soil cores under the alpine *Kobresia* meadow (AKM), interspace meadow patches of *P. fruticosa* shrubs (MPFS), *P. fruticosa* shrubs (PFS) and *A. sphaerocephala* shrubs (ASS)

(Fig. 4b) due to the invasion of *P. fruticosa* shrub roots. For the soils under the *P. fruticosa* shrubs, the roots were coarser and exhibited more complex shapes than the roots of the interspace patches of *P. fruticosa* shrubs (Fig. 4c). The coarse shrub taproots under the *P. fruticosa* shrubs were concentrated within the surface soil (0–100 mm), the distribution of the roots decreased with depth (Fig. 5c). For the soils under the *A. sphaerocephala* shrubs, the shrub roots were observed clearly in the soil cores (Fig. 4d), and the roots were coarser than those under the *P. fruticosa* shrubs. The shrub roots under the *A. sphaerocephala* shrubs were concentrated throughout the surface layer (0–100 mm); in contrast to the other three sites, a crest was observed from 50 to 100 mm (Fig. 5d).

The 3-D characteristics of the root networks determined from the CT images are shown in Table 1 for all of the sampled columns. The root volume density of the *P. fruticosa* shrub was 3 times greater than that of the interspace meadow patches of *P. fruticosa* shrubs ($0.0147 \text{ m}^3 \text{ m}^{-3}$) and approximately 20 times greater than that of the alpine *Kobresia* meadow ($0.0023 \text{ m}^3 \text{ m}^{-3}$). The volume density, number density,

surface area density, length density, node density, branch density, mean root size and tortuosity of the roots of the *P. fruticosa* shrubs were the greatest among all of the soil cores. Furthermore, similar to the volume density, significant differences were observed in the surface area densities, node densities and branch densities among all of the sites ($p < 0.05$). In addition, we found an increase in the volume fraction, number, mean size and radius of the roots with an increase in the shrub invasion (*P. fruticosa* shrubs and *A. sphaerocephala* shrubs > interspace meadow patches of *P. fruticosa* shrubs > alpine *Kobresia* meadow). The mean size and equivalent radius of the roots of the *A. sphaerocephala* shrubs were the greatest due to the growth of the shrubs at the sampling sites. Among all of the sites, the most common root size (0–100 mm^3) accounted for more than 40% of the total while the larger roots (> 1000 mm^3) accounted for approximately 10% of all counted roots (Fig. 6).

3.3 Visualisation and Quantification of the 3-D Macropore Networks

The 3-D visualisations of the soil macropore networks at each of four sites are shown in Fig. 7, in which the blue parts represent macropores. Figure 5 e–h illustrates the distribution of the macroporosity along the soil column for each of the four sites. The macropore characteristics were distinctly different among the four sites. Different types of macropores, including continuous macropores, horizontal and vertical macropores, point-like pores and cracks, were also observed.

For the soils under the alpine *Kobresia* meadow, discontinuous macropores and point-like pores were randomly distributed throughout the surface layer (0–100 mm) (Fig. 7a), and the macroporosity was the smallest among all of the sampling sites (Fig. 5e). For the soils under the interspace meadow patches of *P. fruticosa* shrubs, many continuous macropores

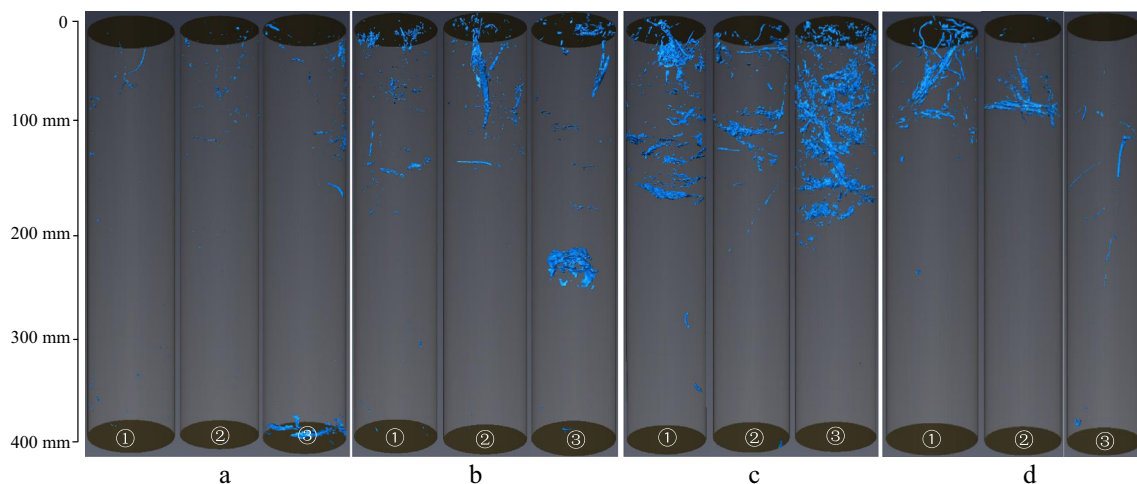


Fig. 7 Three-dimensional visualisations of the soil macropore networks in the soil cores under the alpine *Kobresia* meadow (a), interspace meadow patches of *P. fruticosa* shrubs (b), *P. fruticosa* shrubs (c) and *A. sphaerocephala* shrubs (d)

Table 2 Quantified 3-D characteristics of the macropore networks for all of the soil cores as determined from the CT images

	Macroporosity ($\text{m}^3 \text{ m}^{-3}$)	Macropore number density ($\times 10^5$) (no. m^{-3})	Surface area density ($\text{m}^2 \text{ m}^{-3}$)	Length density (km m^{-3})	Node density ($\times 10^5$) (no. m^{-3})	Branch density (no. m^{-3})	Mean macropore length (mm)	Mean branch length (mm)	Mean macropore size (mm^3)	Hydraulic radius (mm)	Mean coordination numbe	Tortuosity	Mean angle ($^\circ$)
AKM1	0.0002	1.99	0.0009	0.21	0.96	0.59	3.71	3.57	0.91	0.52	1.24	2.41	59.67
AKM2	0.0003	4.05	0.0014	0.34	2.00	1.19	2.84	2.89	0.68	0.50	1.19	0.61	46.28
AKM3	0.0011	3.04	0.0035	0.88	3.75	3.21	2.75	2.72	3.67	0.64	1.71	1.19	64.51
Mean	0.0005 (± 0.0002) ^a	3.02 (± 0.34)	0.0019 (± 0.0005)	0.48 (± 0.12) ^a	2.23 (± 0.47) ^a	1.66 (± 0.46) ^a	3.1 (± 0.18)	3.06 (± 0.15)	1.75 (± 0.55)	0.55 (± 0.02)	1.38 (± 0.1)	1.4 (± 0.31)	56.82 (± 3.15)
MPFS1	0.0010	5.82	0.0038	0.88	4.45	3.36	2.70	2.61	1.69	0.60	1.51	1.20	59.58
MPFS2	0.0016	3.76	0.0039	0.99	3.56	2.96	3.56	3.33	4.25	0.72	1.66	1.17	53.43
MPFS3	0.0031	2.36	0.0061	1.22	4.18	4.04	2.62	3.01	12.99	0.90	1.93	1.21	64.45
Mean	0.0019 (± 0.0004) ^a	3.98 (± 0.58)	0.0046 (± 0.0004)	1.03 (± 0.06) ^a	4.07 (± 0.15) ^a	3.45 (± 0.18) ^a	2.96 (± 0.17)	2.99 (± 0.12)	6.31 (± 1.97)	0.74 (± 0.05)	1.7 (± 0.07)	1.19 (± 0.01)	59.16 (± 1.84)
PFS1	0.0044	4.23	0.0106	2.34	8.02	7.54	3.34	3.10	10.31	0.77	1.88	0.56	58.72
PFS2	0.0046	11.38	0.0142	3.70	14.38	13.51	3.13	2.74	4.06	0.63	1.88	1.18	63.02
PFS3	0.0133	3.19	0.0887	3.81	12.80	10.84	3.28	3.51	1.53	1.05	1.69	0.86	53.47
Mean	0.0074 (± 0.0017) ^b	6.27 (± 1.49)	0.0379 (± 0.0147)	3.28 (± 0.27) ^b	11.73 (± 1.1) ^b	10.63 (± 1.00) ^b	3.25 (± 0.04)	3.12 (± 0.13)	5.3 (± 1.51)	0.82 (± 0.07)	1.82 (± 0.04)	0.86 (± 0.1)	58.4 (± 1.59)
ASS1	0.0015	1.54	0.0041	0.82	2.29	1.83	4.81	4.48	9.55	0.76	1.60	1.17	48.87
ASS2	0.0011	1.96	0.0033	0.80	2.98	2.93	2.75	2.71	5.83	0.68	1.97	1.21	60.41
ASS3	0.0003	0.29	0.0265	0.15	0.33	0.22	7.56	6.53	9.59	0.79	1.36	1.21	60.41
Mean	0.001 (± 0.0002) ^a	1.27 (± 0.29)	0.0113 (± 0.0044)	0.59 (± 0.13) ^a	1.86 (± 0.46) ^a	1.66 (± 0.46) ^a	5.04 (± 0.8)	4.57 (± 0.64)	8.32 (± 0.72)	0.74 (± 0.02)	1.64 (± 0.10)	1.19 (± 0.01)	56.56 (± 2.22)

AKM, MPFS, PFS and ASS represent the samples of alpine *Kobresia* meadow, interspace meadow patches of *P. fruticosa* shrubs and *A. sphaerocephala* shrubs, respectively. The numbers in parentheses are standard errors of the mean. The letters after the parentheses indicate the significance test of the mean difference among the four groups of soil columns at $p < 0.05$. Mean comparison by using the Tukey HSD test

and cracks were observed in the surface layer (0–150 mm) (Fig. 7b). Moreover, the macroporosity, which was larger than that of the alpine *Kobresia* meadow, decreased with an increase in the depth (0–150 mm) (Fig. 5f). For the soils under the *P. fruticosa* shrubs, the continuous macropores, cracks and point-like pores were larger than those at the other three sites, and most of them were distributed around the shrub roots (Fig. 7c). The macroporosity in the surface soil (0–150 mm) was the largest, and it generally decreased with an increase in the soil depth (Fig. 5g). For the soils under the *A. sphaerocephala* shrubs, the soil macropores were continuously distributed around the shrub roots (Fig. 7d). Furthermore, the macroporosity in the surface layer (0–100 mm) was the largest, and it dwindled to approximately zero below a depth of 100 mm (Fig. 5h).

The soil macropore characteristics of all of the soil cores are shown in Table 2. For the soils under the *P. fruticosa* shrubs, the macroporosity was high ($0.0074 \text{ mm}^3 \text{ mm}^{-3}$), which was over 3.5 times greater than that of the interspace meadow patches of *P. fruticosa* shrubs ($0.0019 \text{ m}^3 \text{ m}^{-3}$) and approximately 15 times greater than that of the alpine *Kobresia* meadow ($0.0005 \text{ m}^3 \text{ m}^{-3}$). Similar to the root characteristics, the macroporosity, number density, surface area density, length density, node density, branch density, hydraulic radius and mean coordination number of the macropores in the soils beneath the *P. fruticosa* shrubs were the greatest among the four sites. Meanwhile, there were significant differences in the macroporosity, length density, node density and branch density among the four sites ($p < 0.05$). In addition, obvious increases in the macroporosity, surface area density and hydraulic radius were observed with an increase in shrub invasion (*P. fruticosa* shrub and *A. sphaerocephala* shrub > interspace meadow patches of *P. fruticosa* shrub > alpine *Kobresia* meadow). The mean macropore length (5.04 mm) for the *A. sphaerocephala* shrub was the longest among the four sites, indicating that the interconnectivity of the macropores was the highest. Similar to the results of Luo et al. (2010a), the mean hydraulic radius and the tortuosity of the macropores for the *P. fruticosa* shrub were approximately 0.7 mm and 0.86–1.4, respectively. The mean macropore angles of the interspace meadow patches of *P. fruticosa* shrub were the greatest among all of the sites, indicating the presence of macropores that are more horizontal, and the macropore branch density fluctuated between 1.66 and $10.63 \text{ m}^3 \text{ m}^{-3}$, which was similar to the results of Kuka et al. (2013).

The relative frequency distributions of the macropore sizes at all of the sites are shown in Fig. 8. The macroporosity alone could be used as an indicator of the various characteristics in this study, which is similar to the findings of Katuwal et al. (2015). Moreover, the most common macropore volumes were in the range of 0–100 mm^3 , which accounted for more than 80% of all counted macropores (Fig. 8), while the size

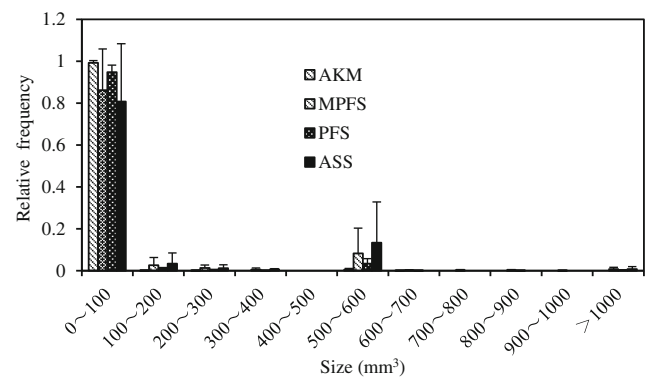


Fig. 8 Cumulative relative frequency distributions of the macropore size in the soil cores under the AKM (alpine *Kobresia* meadow), MPFS (interspace meadow patches of *P. fruticosa* shrubs), PFS (*P. fruticosa* shrubs) and ASS (*A. sphaerocephala* shrubs)

frequency of 500–600 mm^3 accounted for approximately 20% of all counted macropores.

3.4 Relationships Between the Roots and Soil Macropores

The results of a correlation analysis between the soil macropore characteristics and the root characteristics among the four sites are shown in Table 3. There were significant positive correlations ($p < 0.01$) between the volume density, branch density and mean angle of the roots and those of the macropores for all sites. Furthermore, there were also significant positive correlations between the root volume density and the surface area density, length density, node density, branch density and hydraulic radius of the macropores. Therefore, the surface area density, length density, node density, branch density and hydraulic radius of the macropores increased with an increase in the root volume density.

4 Discussion

The results of Kuka et al. (2013) showed the confidence in the suitability of the X-ray micro-CT methodology for root evaluation in undisturbed soil cores. However, the question that needs to be verified in this study is whether these virtual images of medical X-ray CT scanner really reflect the actual root distribution in the soil cores. Heeraman et al. (1997) reported that the length of root per unit volume of soil was greater when measured via CT. However, there was also a strong relationship between the two methods with respect to the root length. The results obtained using the non-destructive approach coincided well with the actual segment dimensions. Both the soil macropores and the roots were undisturbed during the X-ray CT scanning approach, resulting in fewer errors compared with the tedious handwashing method, which is subject to greater human error. These results provide confidence in the

Table 3 Correlation analysis (Pearson's correlation) between the quantified characteristics of roots and macropores derived from the CT scans of soil samples

	Macropore	Macropore number density	Surface area density	Length density	Node density	Branch density	Mean macropore length	Mean branch length	Mean macropore size	Hydraulic radius	Mean coordination number	Tortuosity	Mean angle
Root volume density	0.86 ^{***}	0.19	0.73 ^{***}	0.81 ^{***}	0.74 ^{***}	0.74 ^{***}	-0.09	-0.06	0.17	0.714 ^{***}	0.543 [*]	-0.45	0.00
Root number density	-0.09	0.11	0.07	-0.03	-0.02	-0.02	0.55 [*]	0.50 [*]	0.19	0.16	-0.04	0.01	-0.02
Surface area density	0.61 [*]	0.54 [*]	0.49	0.69 ^{***}	0.72 ^{***}	0.69 ^{***}	-0.34	-0.35	-0.31	0.17	0.15	-0.67 ^{***}	-0.16
Length density	-0.36	-0.14	-0.29	-0.43	-0.38	-0.40	-0.19	-0.18	-0.54 [*]	-0.62 [*]	-0.62 [*]	0.40	-0.11
Node density	0.47	0.57 [*]	0.36	0.57 [*]	0.60 [*]	0.57 [*]	-0.30	-0.32	-0.60 [*]	-0.13	-0.21	-0.02	-0.14
Branch density	0.64 [*]	0.61 [*]	0.51 [*]	0.73 ^{***}	0.76 ^{***}	0.72 ^{***}	-0.29	-0.30	-0.54 [*]	0.06	-0.05	-0.13	-0.16
Mean root length	-0.11	0.16	-0.20	0.09	0.10	0.16	-0.08	-0.17	0.23	-0.03	0.55 [*]	-0.18	0.30
Mean branch length	-0.13	-0.23	-0.12	-0.15	-0.16	-0.13	-0.17	-0.18	0.05	-0.06	0.38	0.03	0.17
Mean root size	0.13	0.03	-0.07	0.23	0.20	0.23	-0.10	-0.14	0.42	0.13	0.38	-0.45	0.05
Mean radius	-0.13	0.03	-0.24	0.09	0.07	0.15	0.16	0.11	0.78 ^{***}	0.17	0.57 [*]	-0.27	0.48
Mean coordination number	0.28	-0.02	0.16	0.29	0.24	0.26	-0.17	-0.18	0.13	0.29	.606 [*]	-0.23	-0.13
Tortuosity	0.21	-0.09	0.08	0.23	0.18	0.21	-0.19	-0.21	0.21	0.17	0.51 [*]	-0.34	0.07
Mean angle	-0.22	0.06	-0.26	-0.15	-0.11	-0.07	-0.13	-0.11	0.40	0.07	0.26	0.09	0.65 ^{***}

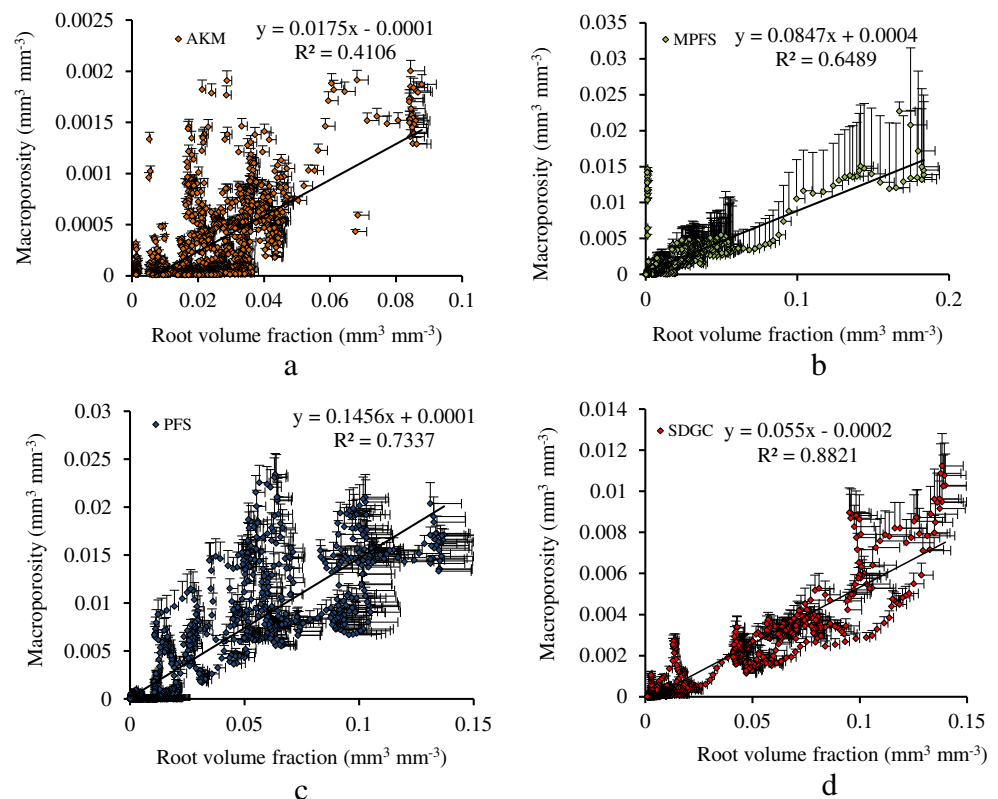
Probability values in parentheses are delineated by two significance levels (^{***} < 0.01, ^{*} < 0.05)

suitability of the medical X-ray CT methodology for evaluating the root system in an undisturbed soil column. The relationships between the root volume density and the macroporosity from the CT slices are shown in Fig. 9. A previous study reported that the high number of macropores under the enclosed shrubland was closely associated with high root density (Hu et al. 2019). The root volume density was significantly correlated with the macroporosity with $R^2 = 0.88$ for the *A. sphaerocephala* shrubs, $R^2 = 0.73$ for the *P. fruticosa* shrubs, $R^2 = 0.65$ for the interspace meadow patches of *P. fruticosa* shrubs and $R^2 = 0.41$ for the alpine *Kobresia* meadow. In addition, the root branch density was significantly correlated ($p < 0.01$) with the macropore branch density (Table 3). Therefore, these findings support our hypothesis that the relatively large number of macropores in the soil under the *P. fruticosa* shrubs could be attributed to root development (Hu et al. 2016; Hu et al. 2018). Meanwhile, the roots had little effect on the soil macropores in the alpine *Kobresia* meadow.

The results showed that there were different types of roots and macropores and that the distributions of roots and macropores were distinctly different among all of the sites. Hu et al. (2016) found that the large number of macropores in the soil under *P. fruticosa* shrubs could be attributed to greater root development, although there were no experimental data to verify their conclusion. The macropores formed by roots were highly continuous and round in shape (Luo et al.

2010a), and most of the macropores were distributed in the rhizosphere (Aravena et al. 2011). Previous study indicated that roots were concentrated in the 0–250 mm soil layer in a grazed shrubland in northern China (Hu et al. 2019). However, the results of this study suggest that coarse shrub roots and continuous macropores were concentrated in the surface soil (0–150 mm) under the *P. fruticosa* shrubs, which was consistent with the profile under the *P. fruticosa* shrubs (Fig. 10c). Due to the invasion of *P. fruticosa* shrub roots into the surface soil beneath the interspace meadow patches of *P. fruticosa* shrubs (Fig. 10b), numerous shrub roots and macropores were distributed horizontally throughout the surface soil (0–150 mm) beneath the interspace meadow patches of *P. fruticosa* shrubs. The results of Hu et al. (2015, 2018) suggested that the macroporosity of the soil under shrubs was greater than that of the soil under grassland, and that macropores are mainly formed by developed shrub root systems. Similarly, the macroporosity of the *P. fruticosa* shrubs was significantly greater than that of the interspace meadow patches of *P. fruticosa* shrubs. Moreover, the macroporosity of the *P. fruticosa* shrubs was greater than that of the alpine *Kobresia* meadow, which was consistent with previous findings. Macropores formed by root channels or decomposed root residues accounted for some of the macropore networks at each site. A previous study reported that soil macropores were highly connected and round, with their size generally decreasing with depth, which could contribute to root channels (Luo et al.

Fig. 9 Correlation between the macroporosity and root volume fraction for the soils under the AKM (alpine *Kobresia* meadow) (a), MPFS (interspace meadow patches of *P. fruticosa* shrubs) (b), PFS (*P. fruticosa* shrubs) (c) and ASS (*A. sphaerocephala* shrubs) (d)



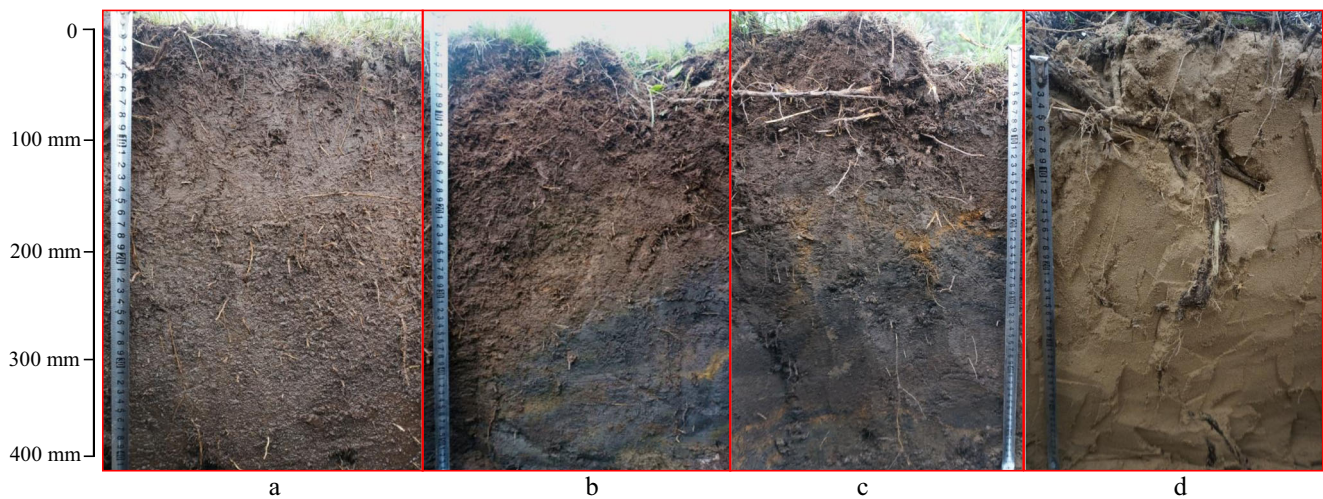


Fig. 10 Soil profiles of the alpine *Kobresia* meadow (a), interspace meadow patches of *P. fruticosa* shrubs (b), *P. fruticosa* shrubs (c) and *A. sphaerocephala* shrubs (d)

2010a). The 3-D visualisation provided in this study suggested that continuous macropores were distributed around the coarse shrub taproots of the *A. sphaerocephala* shrubs, the *P. fruticosa* shrubs and the interspace meadow patches of *P. fruticosa* shrubs. Significant positive correlations between the volume density of the roots and the macroporosity indicate that coarser, larger roots such as shrub roots provide more positive feedback to the development of macropores.

According to the observed profile beneath the alpine *Kobresia* meadow (Fig. 10a), the ‘mattic epipedon’ layer was mainly found in the soil surface layer (0–100 mm). Previous studies indicated that plant roots are dense and compressed within the mattic epipedon layer (Lu et al. 2006; Pang et al. 2009). Moreover, the abundant content of organic matter in the mattic epipedon soil ($203 \times 10^3 \text{ mg kg}^{-1}$) would increase nutrient availability and stimulate root development

(Table 4), which is consistent with a previous study (Martínez et al. 2018). Bulk density is increased with increased relative coarse particles in a soil volume, which alters root growth (Hosseini et al. 2016). Fine herb roots were distributed throughout the entire soil core from the alpine *Kobresia* meadow, but the macropores were concentrated throughout the top layer (0–100 mm). Moreover, the macropores of the alpine *Kobresia* meadow were discontinuous. Small, randomly and discontinuously distributed macropores and cracks were also observed in the soil samples from the *P. fruticosa* shrubs and the interspace meadow patches of *P. fruticosa* shrubs. Previous studies indicated that the freezing and thawing of soils have negative effects (i.e. a reduction) on the macroporosity, thickness and specific surface area of macropores (Starkloff et al. 2017). Hence, most of the randomly and discontinuously distributed macropores

Table 4 Descriptions of the sample sites and basic soil physicochemical properties

Samples	Altitude (m)	Vegetation	Depth (mm)	Organic matter (10^3 mg kg^{-1})	Total phosphorus (10^3 mg kg^{-1})	Total nitrogen (10^3 mg kg^{-1})	Sand (%)	Silt (%)	Clay (%)	Bulk density (g cm^{-3})
Alpine <i>Kobresia</i> meadow	3530	<i>Kobresia pygmaea</i> <i>Saussurea pulchra</i> , <i>Polygonum viviparum</i>	0–100	203	1.04	11.80	76.53	19.34	4.14	0.48
			100–200	216	0.98	8.95	50.43	42.73	6.85	0.49
			200–400	165	0.84	7.42	70.30	20.00	9.70	0.48
Interspace meadow patches of <i>P.</i> <i>fruticosa</i> shrubs	3564	<i>Carex moorcroftii</i> and <i>P. saundersiana</i>	0–100	204	1.03	10.70	82.26	14.49	3.25	0.66
			100–200	173	1.29	10.80	56.26	35.93	7.81	0.75
			200–400	179	0.91	7.23	54.98	37.06	7.96	0.90
<i>P. fruticosa</i> shrubs	3566	<i>P. fruticosa</i> , <i>Carex</i> <i>moorcroftii</i>	0–100	148	0.94	6.95	45.38	45.47	9.15	0.47
			100–200	164	1.01	7.22	57.57	35.36	7.07	0.57
			200–400	95	0.51	3.84	57.17	34.23	8.60	0.70
<i>A. sphaerocephala</i> shrubs	3226	<i>A. sphaerocephala</i>	0–100	2.09	0.24	0.16	96.14	3.57	0.28	1.56
			100–200	0.69	0.15	0.13	97.46	2.48	0.06	1.38
			200–400	0.72	0.22	0.13	98.09	1.91	0.00	1.54

were interaggregate macropores formed by freezing and thawing (Luo et al. 2010a).

5 Conclusions

The results revealed that the root volume density of the *P. fruticosa* shrubs was greater than that of the interspace meadow patches of *P. fruticosa* shrubs ($0.0147 \text{ m}^3 \text{ m}^{-3}$) and approximately 20 times greater than that of the alpine *Kobresia* meadow ($0.0023 \text{ m}^3 \text{ m}^{-3}$). The macroporosity ($0.0074 \text{ mm}^3 \text{ mm}^{-3}$) and the root volume density ($0.045 \text{ mm}^3 \text{ mm}^{-3}$) of the soils under the *P. fruticosa* shrubs patches were significantly greater than those in the soils under the other sites. In addition, the characteristics of both roots and macropores (i.e. the volume density and number density) increased with an increase in shrub invasion (*P. fruticosa* shrubs > interspace meadow patches of *P. fruticosa* shrubs > alpine *Kobresia* meadow). This study clearly demonstrates that the root distribution exhibits positive effects on the development of soil macropores. There were significant positive correlations ($p < 0.01$) between the volume density, branch density and mean angle of the roots and those of the macropores for all sites. The root volume density was the most important factor that influenced the quantified characteristics and distributions of the macropores. The soil macropores under the mattic epipedon layer were smaller and less continuous, which could be attributable to freezing and thawing. Our findings provide valuable insights into understanding the formation of soil macropores in cold, high-altitude regions.

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Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interest.

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